

2020 .

Q1. (a).

$$\text{Buchanan Galleries} : L_{Bg} = 55^{\circ}51'51'' = 55.86467^{\circ}$$

$$\lambda_{Bg} = 4^{\circ}15'11'' = -4.253056^{\circ}$$

$$\text{Charing Cross} : L_{Cc} = 55^{\circ}51'58'' = 55.86611^{\circ}$$

$$\lambda_{Cc} = 4^{\circ}16'14'' = -4.270556^{\circ}$$

The radius of curvature at each end is

$$R_E(L_{Bg}) = \frac{R_0}{\sqrt{1 - e^2 \sin^2 L_{Bg}}} = 6,392,813.66 \text{ m}$$

$$R_E(L_{Cc}) = \frac{R_0}{\sqrt{1 - e^2 \sin^2 L_{Cc}}} = 6,392,814.34 \text{ m}$$

Buchanan Gallery :

$$x_{eb}^e = R_E(L_{Bg}) \cdot \cos L_{Bg} \cdot \cos \lambda_{Bg} = 3,577,441.79 \text{ m}$$

$$y_{eb}^e = R_E(L_{Bg}) \cdot \cos L_{Bg} \cdot \sin \lambda_{Bg} = -266,045.43 \text{ m}$$

$$z_{eb}^e = (1 - e^2) R_E(L_{Bg}) \sin L_{Bg} = 5,255,970.30 \text{ m}$$

Charing Cross :

$$x_{cb}^e = R_E(L_{Cc}) \cdot \cos L_{Cc} \cdot \cos \lambda_{Cc} = 3,577,231.33 \text{ m}$$

$$y_{cb}^e = R_E(L_{Cc}) \cdot \cos L_{Cc} \cdot \sin \lambda_{Cc} = -267,124.73 \text{ m}$$

$$z_{cb}^e = (1 - e^2) \cdot R_E(L_{Cc}) \cdot \sin L_{Cc} = 5,256,091.76 \text{ m}$$

$$\text{dist}(\underline{a}, \underline{b}) = \|\underline{b} - \underline{a}\| = 1116.9 \text{ m}$$

$$(b). \underline{r}_{ap}^a = \underline{r}_{ep}^a - \underline{r}_{ea}^a = \underline{r}_{eb}^a + \underline{r}_{bp}^a - \underline{r}_{ea}^a$$

$$= C_e^a \underline{r}_{eb}^e + C_b^a \underline{r}_{bp}^b - C_e^a \underline{r}_{ea}^e$$

$$\Rightarrow \dot{\underline{r}}_{ap}^a = \dot{C}_e^a \underline{r}_{eb}^e + C_e^a \dot{\underline{r}}_{eb}^e + \dot{C}_b^a \underline{r}_{bp}^b + C_b^a \dot{\underline{r}}_{bp}^b - \dot{C}_e^a \underline{r}_{ea}^e - C_e^a \dot{\underline{r}}_{ea}^e$$

$$= \Omega_{ea}^a C_e^a \underline{r}_{eb}^e + C_e^a \underline{V}_{eb}^e - \Omega_{ba}^a C_b^a \underline{r}_{bp}^b + C_b^a \underline{V}_{bp}^b$$

$$+ \Omega_{ea}^a C_e^a \underline{r}_{ea}^e - C_e^a \underline{V}_{ea}^e$$

$$= \underline{V}_{eb}^a - \underline{V}_{ea}^a - \Omega_{ba}^a C_b^a \underline{r}_{bp}^b - \Omega_{ea}^a (\underline{r}_{eb}^a - \underline{r}_{ea}^a) + \underline{V}_{bp}^a$$

$$\underline{\omega}_{ba}^a = \underline{\omega}_{eb}^a - \underline{\omega}_{ea}^a \Rightarrow \Omega_{ba}^a = \Omega_{eb}^a - \Omega_{ea}^a$$

$$\underline{V}_{bp}^a = 0$$

$$\Rightarrow \dot{\underline{r}}_{ap}^a = \underline{V}_{eb}^a - \underline{V}_{ea}^a - \Omega_{eb}^a C_b^a \underline{r}_{bp}^b + \Omega_{ea}^a C_b^a \underline{r}_{bp}^b - \Omega_{ea}^a (\underline{r}_{eb}^a - \underline{r}_{ea}^a) + \underline{V}_{bp}^a$$

$$= \underline{V}_{eb}^a - \underline{V}_{ea}^a - C_b^a \Omega_{eb}^a \underline{r}_{bp}^b + \Omega_{ea}^a (\underline{r}_{eb}^a - \underline{r}_{ea}^a + C_b^a \underline{r}_{bp}^b)$$

Q2.

Step1: Transform specific force :

$$\underline{f}_{ib}^a = C_b^a \underline{f}_{ib}^b = \begin{bmatrix} -0.016386 \\ 0.010489 \\ -0.047130 \end{bmatrix}$$

$$\text{Step2: Gravity} = g_b^a(Lb, h_s) = 9.7803253 \frac{(1 + 0.00193185 \sin^2 L)}{\sqrt{1 - e^2 \sin^2 L}}$$

$$\begin{bmatrix} 0 \\ 0 \\ 9.815817 \end{bmatrix} = 9.815817$$

Step 3: Ω_{en}^n and Ω_{ie}^e

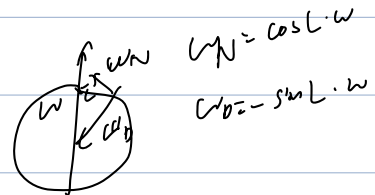
$$R_N(L) = \frac{R_0(1-e^2)}{(1-e^2\sin^2 L)^{3/2}} = 6,379,240,31 \text{ m}$$

$$R_E(L) = \frac{R_0}{\sqrt{1-e^2\sin^2 L}} = 6,392,818.68 \text{ m}$$

$$\underline{\omega}_{en}^n = \begin{bmatrix} v_{eb,E}^n / (R_E(L_b) + h_b) \\ -v_{eb,N}^n / (R_N(L_b) + h_b) \\ -v_{eb,E}^n \cdot \tan(L_b) / (R_E(L_b) + h_b) \end{bmatrix} = \begin{bmatrix} 0 \\ -3.759816 \times 10^{-5} \\ 0 \end{bmatrix}$$

$$\Rightarrow \hat{\Omega}_{en} = \begin{bmatrix} 0 & 0 & -3.759816 \times 10^{-5} \\ 0 & 0 & 0 \\ 3.759816 \times 10^{-5} & 0 & 0 \end{bmatrix}$$

$$\Omega_{ie}^e = \begin{bmatrix} 0 & 6.036783 \times 10^{-5} & 0 \\ -6.036783 \times 10^{-5} & 0 & -4.090509 \times 10^{-5} \\ 0 & 4.090509 \times 10^{-5} & 0 \end{bmatrix}$$



Step 4: $\dot{v}_{eb}^n = \underline{f}_{ib}^n + \underline{g}_b^n(L_b, h_b) - (\Omega_{en}^n + 2\Omega_{ie}^e) \cdot v_{eb}^n$

$$= \begin{bmatrix} -0.016386 \\ 0.010489 \\ 9.768687 \end{bmatrix} -$$

$$= \begin{bmatrix} -0.016386 \\ 0.039466 \\ 9.759663 \end{bmatrix}$$

Step 5: Using the trapezoidal rule.

$$\vec{v}_b^n = \vec{v}_b^n(t_{k-1}) + \Delta t \cdot \frac{\vec{v}_b^n(t_k) + \vec{v}_b^n(t_{k-1})}{2} = \begin{bmatrix} 239.999181 \\ 0.001973 \\ 0.487983 \end{bmatrix} \text{ m/s}$$

Step 6: $\dot{L}_b$, $\dot{\lambda}_b$, $\dot{h}_b$

$$\dot{L}_b(t_{k-1}) = \frac{v_{b,n}^n(t_{k-1})}{R_W(L_b) + h_b} = 3.759816 \times 10^{-5} \text{ rad/s}$$

$$\dot{L}_b(t_k) = \frac{v_{b,n}^n(t_k)}{R_W(L_b) + h_b} = 3.759804 \times 10^{-5} \text{ rad/s}$$

$$\dot{\lambda}_b(t_{k-1}) = \frac{v_{b,e}^n(t_{k-1})}{(R_W(L_b) + h_b) \cos L_b} = 0 \text{ rad/s}$$

$$\dot{\lambda}_b(t_k) = \frac{v_{b,e}^n(t_k)}{(R_W(L_b) + h_b) \cos L_b} = 5.51 \times 10^{-10} \text{ rad/s}$$

$$\dot{h}_b(t_{k-1}) = -v_{b,r}^n(t_{k-1}) = 0$$

$$\dot{h}_b(t_k) = -v_{b,r}^n(t_k) = -0.487983 \text{ m/s}$$

Step 7: $L_b = L_b(t_{k-1}) + \frac{\Delta t}{2} [\dot{L}_b(t_{k-1}) + \dot{L}_b(t_k)] \frac{180}{\pi} ?$

$$= 55.878590^\circ ?$$

$$55.878802^\circ ?$$

$$\lambda_b = \lambda_b(t_{k-1}) + \frac{\Delta t}{2} [\dot{\lambda}_b(t_{k-1}) + \dot{\lambda}_b(t_k)]$$

$$= -4.676749$$

$$h_5 = - \dots$$

Q3. (b).

Time Update:

$$\begin{aligned} \hat{x}_k^- &= \Phi_{k-1} \hat{x}_{k-1}^+ = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0.1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.40 \\ 1.50 \\ -0.50 \end{bmatrix} \\ &= \begin{bmatrix} -0.50 \\ 0.55 \\ -0.50 \end{bmatrix} \end{aligned}$$

$$P_k^- = \Phi_{k-1} P_{k-1}^+ \Phi_{k-1}^T + Q$$

$$= \begin{bmatrix} 13.500 & 4.550 & 3.500 \\ 7.070 & 13.809 & 7.070 \\ 3.500 & 4.550 & 5.500 \end{bmatrix}$$

Measurement update:

$$\begin{aligned} K_k &= P_k^- H_k^T [H_k P_k^- H_k^T + R_k]^{-1} \\ &= \begin{bmatrix} 0.69107 \\ 0.448276 \\ 0.691078 \end{bmatrix} \quad (3.8190) \end{aligned}$$

$$\hat{x}_{k+1} = \hat{x}_k + k_1 (z_1 - H \hat{x}_k)$$

$$= \begin{bmatrix} 0.329292 \\ 1.709137 \\ 0.329794 \end{bmatrix}$$

Con.

True ranges:

$$B_1 = \sqrt{(300-30)^2 + (250-30)^2} = 348.281495$$

$$B_2 = \sqrt{(-50-30)^2 + (175-30)^2} = 165.604952$$

$$B_3 = \sqrt{(20-30)^2 + (-90-30)^2} = 120.415946$$

Iteration 1:

Range estimates for $\underline{x} = (0, 0)$

$$R_1 = \sqrt{(300-0)^2 + (250-0)^2} = 390.512484$$

$$R_2 = \sqrt{(-50-0)^2 + (175-0)^2} = 182.002747$$

$$R_3 = \sqrt{(20-0)^2 + (-90-0)^2} = 92.195445$$

Construct Jacobian:

$$H = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{0-300}{390.512484} & \frac{0-250}{390.512484} \\ \frac{0+50}{182.002747} & \frac{0-175}{182.002747} \\ \frac{0-20}{92.195445} & \frac{0+90}{92.195445} \end{bmatrix} = \begin{bmatrix} -0.768221 & -0.640184 \\ 0.27921 & -0.961524 \\ -0.216930 & 0.976187 \end{bmatrix}$$

Get range error:

$$\delta R = \begin{bmatrix} B_1 - R_1 \\ B_2 - R_2 \end{bmatrix} = \begin{bmatrix} -42.230989 \\ -16.397795 \end{bmatrix}$$

$$\begin{bmatrix} B_3 - R_3 \end{bmatrix} = \begin{bmatrix} 28.220501 \end{bmatrix}$$

Position update: $A=H$ $B=R$

$$\delta u = (H^T H)^{-1} H^T \{R$$

$$= \begin{bmatrix} 29.929662 \\ 30.569248 \end{bmatrix}$$

The new position estimate is:

$$u_1 = u_0 + \delta u$$

$$= (29.929662, 30.569248)$$

$$\|\delta u\| = 42.765163 > 0.1$$

Iteration 2:

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